

AAPG2026	VIDI		PRCE
Coordinated by	Lina ROJAS-BARAHONA	48 months	927k€
CE23 – Artificial Intelligence and Data Science			

VIDI: Visual Dialogue for Visually Impaired Navigation Guidance

Consortium: **Orange** (coordinator), **ENPC**, **Inria**, **IRIT**

Summary

The project Visual Dialogue for visually impaired navigation guidance (VIDI, or “I saw” in Latin) aims to advance artificial intelligence (AI) systems to assist visually-impaired people in vision-and-language navigation through situated multi-modal dialogue. In this project, we will leverage audio-enabled wearable cameras to facilitate such dialogues; for example, answering questions, warning users about changes in the environment, and guiding them to target objects. Despite the potential impact on assistive technologies, there is no existing AI solution for long-term guidance for the visually impaired. Our goal is to provide language feedback to the camera wearer by building vision-language models (VLMs) capable of navigating and engaging in conversation given multi-modal inputs such as the transcribed dialogue and the egocentric video stream. We will open-source the software outcomes from the project, as well as the data collection with visually-impaired users, in compliance with data governance regulations.

Summary table of people involved in the project:

Partner	Name	First name	Current position	Role & responsibilities	Involvement (pers.month)
Orange	ROJAS-BARAHONA	Lina	Researcher	Coordinator WP0, WP1, WP3, WP4, WP5	22 p.m
Orange	URVOY	Tanguy	Researcher	Contributor WP2, WP3	16 p.m
Orange	VEYRET	Morgan	Research Engineer	Contributor WP1, WP3, WP4	15.5 p.m
Orange	TARNEC	Christèle	Research Engineer	Contributor WP1, WP5	6 p.m
Orange	BOTHEREL	Valérie	Ergonomist	Contributor WP1, WP4 WP5	10 p.m
LIGM, ENPC	VAROL	Gül	Researcher	Co-PI WP0, WP1, WP2, WP3	6 p.m
Inria Paris	CHEN	Shizhe	Researcher	Co-PI WP0, WP1, WP2	9 p.m
IRIT	SERPA	Antonio	CNRS Engineer	Co-PI WP0, WP1, WP5	15 p.m
IRIT	ORIOLA	Bernard	CNRS Engineer	Co-PI WP1, WP5	15 p.m
Orange	(hiring)	-	Post-doc	WP3, WP4	24 p.m
ENPC	(hiring)	-	PhD	WP3	36 p.m
Inria Paris	(hiring)	-	PhD	WP2	36 p.m
IRIT	(hiring)	-	Engineer	WP1	12 p.m
IRIT	(hiring)	-	Post-doc	WP5	18 p.m
IRIT	(hiring)	-	MS Intern	WP1	6 p.m
IRIT	(hiring)	-	MS Intern	WP5	6 p.m
All					252.5 p.m

Changes with respect to the AAPG2025 proposal (first PRCE project in the complementary list):

- We integrate reviewers comments: hypotheses and valorisation.
- We update the state-of-the-art and our involvement in other collaborative projects.
- We adjust the budget only due to yearly staff cost and, regarding the preproposal, the requested funds increased solely by +0.85% due to material costs: a high-performance computational unit and 4 wearable devices.

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I Proposal's context, positioning and objective(s)

Context and motivation. In France, approximately 1.7 million people are affected by visual impairment, a number expected to double by 2050 as it becomes increasingly prevalent with aging. Recently, AI-driven applications that assist **visually impaired people (VIP)** include smartphone apps (e.g., Seeing AI¹ and OOrion²) and assistive glasses³. However, these technologies face three major limitations that hinder their ability to satisfy the needs of VIP. First, they focus on describing scenes or reading text but do not provide actionable guidance, such as how to navigate to a target destination, a common requirement for VIP. Second, these technologies typically rely on the user capturing an image. However, not only this might be difficult for VIP but also a single image lacks the contextual information needed to handle more complex tasks requiring video (e.g., guiding VIP back to a door entrance that is no longer visible from their current location). Third, existing applications are limited to simple, one-way interactions, generating a single response to one question from the user. This fails to communicate with VIP in a natural and continuous manner. Moreover, expecting the user to initiate with a question ignores critical scenarios where the machine should warn about obstacles, inform of new people in the environment, or simply ask the user to move their head to better capture the scene. For all these reasons, there is an urgent need to **develop AI methods that can more actively interact with VIP and assist them in navigating the 3D world.**

a Objectives and scientific hypotheses

The project '**VI**sual **D**ialogue for visual impaired navigation guidance' (VIDI) aims to provide automated assistance to VIP through audio-enabled wearable cameras (e.g. smart glasses with cameras and microphone, smartphones etc.). Our goal is to build AI models, which can engage in a **situated multimodal dialogue (MMD)** with the VI user given the egocentric video recording, as well as the text inputs. More concretely, we will build vision and language models (VLMs) that can understand and generate text to guide a VI user. In this project, we focus on text generation, however, the speech-recognition and text-to-speech (TTS) conversion can trivially be performed at real-world deployment. We will adopt a co-construction strategy at each stage of the project and an accessibility checklist will ensure every deliverable meets international standards, whenever applicable, and the needs of VIPs.

Figure 1 illustrates a potential real-world scenario. While the scenarios will be determined during our data collection phase, example target capabilities include multimodal dialogue, vision-and-language navigation, obstacle detection, scene description, as well as semantic and visual grounding among others.



Figure 1. Real-world scenario where VIDI helps VIP obtain drinking water in a new environment.

In VIDI we hypothesize that: **H1: user-centered datasets better capture the diversity of VIP's needs;** **H2: generalization, long-term memory and robustness in dynamic environments are key factors for efficient vision-and-language navigation** and **H3: multimodal dialogue systems that integrate long-form video understanding and handle conversational linguistic phenomena⁴ are expected to**

¹https://en.wikipedia.org/wiki/Seeing_AI

²<https://www.oorion.fr/>

³<https://www.nature.com/articles/s41433-023-02842-z>

⁴For example: *mixed initiative* (speakers can interchangeably initiate the conversation, affirm, request or respond), *grounding* (speakers agree in their mutual understanding) and *planning* (speakers select the most effective course of action to accomplish a task).

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increase both task success rate and VIP's satisfaction.

To verify these hypothesis we define the following main objectives :

- **O1: Collect multimodal data** for VIP assistance using wearable devices and satisfying VI needs.
- **O2: Train generalizable vision-and-language navigation (VLN) models** that can follow user demands to find obstacle-free paths to reach target objects in diverse environments.
- **O3: Develop situated multimodal dialogue** capability for VLMs to support VIP assistance.
- **O4: Propose quantitative and qualitative evaluations for dialogue in VIP assistance tasks**, to assess the interaction quality.

To reach these objectives we will address the following scientific challenges:

Challenges in realistic datasets. The main challenge is to use user-centered design methods adapted to VIPs during the needs gathering phase (interviews, focus groups, etc.), the co-construction of realistic activity scenarios and the data collection phase. The objective is to obtain a dataset that is as representative as possible of real dialogue between VIPs and guides and to ensure that the profiles of the VIPs are as representative as possible.

Challenges in vision-and-language navigation. Three key challenges must be addressed to enable models to follow user instructions to navigate in diverse environments. The first difficulty lies in *generalization across environments and instructions*. VIP frequently encounter unfamiliar environments in daily life, containing novel objects and spatial arrangements. User instructions also vary in specificity, ranging from high-level goals from VIP like "find a coffee mug" to detailed step-by-step guidance from others like "go straight until the bank building and turn right". The scarcity of training data, compared to the vast variations of real-world scenarios, further exacerbates the generalization challenge. Second, *maintaining compact yet rich spatial memory* is essential for VLN. Rather than relying solely on current observations or short-term memory, models require a long-term structured memory to support strategic planning for navigation, such as backtracking. Beyond geometric information, semantic cues should also be integrated into memory representations to enhance the association between vision, language and navigation actions. Finally, *ensuring efficient navigation in dynamic environments* is essential for real-world applications. VLN systems must provide responsive guidance to VIP according to environment changes.

Challenges in situated multimodal dialogue. We need to face distinct challenges to be able to train models that can effectively assist VIP. The first one is to enable models *to support long-form video understanding*, surpassing single interactions and static images. The second challenge is *to detect real-time changes in the environment*. To go beyond the boundaries of existing multimodal dialogue systems, a third challenge is *to support conversational linguistic phenomena*⁴. These challenges involve non-trivial tasks: correctly interpreting the changes in the environment and generating appropriate and timely responses to better guide VI individuals.

b Position of the project as it relates to the state of the art

State of the art for VIP assistants. Automatic assistants for VIP have been widely studied, such as answering questions given images captured by VIP [Chen et al., 2022a], creating navigation systems for blind inside buildings [Kuribayashi et al., 2023], and generating audio descriptions for movies [Han et al., 2023a]. Despite this progress, current systems struggle to generalize for navigation in unknown environments (unseen at model training), lack natural, task-oriented dialogue capabilities to assist VIP in 3D space navigation, and fail to capture long-term context. LLMs [Ouyang et al., 2022] and Vision-Language Models (VLMs) [Awais et al., 2025] attempt to improve visual dialogue interactions, however, task-oriented control remains a challenge [Hudeček and Dusek, 2023; Elizabeth et al., 2025; Acikgoz and Oh, Jinoh et al., 2026]. Interactions with text-only LLMs often are verbose and neglect *mixed initiative, grounding* and *planning*⁴. Overconfidence of LLMs may also be unsafe when assisting VIP, and contradicts grounding (recognizing and communicating uncertainty) [Benotti and Blackburn, 2021]. When it comes to integrating visual data, current datasets for VIP, such as VizWiz [Chen et al., 2022a] and Context-Visual Question Answer (VQA) [Naik et al., 2023] considers only static images, although EgoBlind [Xiao et al., 2025] includes videos, all these datasets are limited to simple interactions. Despite offering richer interactions, MMD datasets [Wu and Kottur, Satwik et al., 2023; Zhang et al., 2023b] are not adapted for VIP since they assume users can see their surroundings and videos considered are short (e.g., 2 minutes). This project aims to push the boundaries of current capabilities for VIP assistance in interactive navigation and situated multimodal dialogues.

In the following, we briefly review most recent methods on relevant subjects: vision-and-language navigation, video understanding and task-oriented dialogue. We also summarize our preliminary results.

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Vision-and-Language Navigation (VLN). After decades of focusing on point-based navigation, research [Anderson and Angel X. Chang et al., 2018] has shifted towards navigation guided by high-level goals, such as object categories, images or natural language instructions. This requires multimodal understanding for goal identification and embodied commonsense reasoning for efficient navigation. Current navigation approaches vary primarily in their map representations, exploration strategies, and planning mechanisms. They generally fall into two categories: end-to-end approaches [Ramakrhya et al., 2022; Chen et al., 2023], and modular approaches [Chaplot et al., 2020; Yokoyama et al., 2024; Huang et al., 2023].

The end-to-end methods directly train neural networks to predict actions. To improve generalization in unseen environments, recent methods have begun leveraging LLMs or VLMs. Early LLM-based VLN methods [Zhou et al., 2024] depend heavily on prompt engineering and post-processing of textual outputs, limiting their capacity to leverage rich visual observations. Later approaches employ VLMs for navigation by conditioning low-level action prediction on the full video history [Zhang et al., 2024b, 2025a; Cheng et al., 2025; Zhang et al., 2026; Chu et al., 2026]. While effective in short-horizon settings, this formulation becomes computationally expensive with long histories, suffers from limited spatial reasoning and weak long-term planning, and remains difficult to interpret. In addition, they typically require large-scale datasets to train.

Modular approaches address these limitations by decoupling semantic mapping, goal prediction, and path planning. Existing semantic maps include 2D spatial grids [Chaplot et al., 2020], topological graphs [Chen et al., 2022b], dense point clouds [Peng et al., 2023], and hybrid representations [Gu et al., 2024]. Spatial grids and graphs are compact but semantically limited, while dense point clouds provide richer semantics at higher memory cost. Hybrid maps aim to balance these trade-offs, though the optimal representation remains open. Goal prediction strategies range from reinforcement learning with extrinsic rewards [Chaplot et al., 2020] to behavior cloning with privileged heuristics [Ramakrishnan et al., 2022; Yokoyama et al., 2024]. Path planning typically relies on deterministic planners [Sethian, 1996] or PointNav policies [Wijmans et al., 2020]. Although modular methods are more data-efficient and interpretable, they have limited generalization across diverse environments and instructions.

Recent efforts combine VLMs with maps [Chiang et al., 2024; Zhang et al., 2025b] to leverage the best of both worlds. However, they either rely on pre-built maps or coarse 2D representations that remain insufficient for navigation in 3D world. In this project, we adopt a modular framework to develop explainable, generalizable, and efficient navigation models, with a focus on improving 3D map representations and strengthening 3D VLMs for better semantic and spatial reasoning.

Video understanding. Our project is also relevant to video understanding, in particular video captioning, video question-answering, and works leveraging LLMs and focusing on long-form videos. Captioning seeks to describe the video content with text. There is a large literature on single video captioning [Chen et al., 2024a] often focusing on short video clips, spanning a few seconds [Bain et al., 2021]. *Dense video captioning* [Krishna et al., 2017] aims to both temporally localize and caption different events. *Question answering (QA)*, on the other hand, considers both a video and a textual question as input [Wang et al., 2024]. In this project, we will leverage QA and captioning approaches, but will tackle the more difficult case of *dialogues*, where the past context needs to be integrated.

LLMs have been leveraged in different ways for improving video understanding. A popular approach is to train ‘bridge’ modules between pretrained visual backbones [Radford et al., 2021] and LLMs to build vision-language models (VLMs) that can ingest videos (e.g., Video-Llama [Zhang et al., 2023a], Video-LLaVa [Lin et al., 2023]). Other works have employed LLMs for automatic construction of video datasets [Islam et al., 2024; Argaw et al., 2024; Ventura et al., 2024], tool use [Min et al., 2024], storing memory in video QA [Kahatapitiya et al., 2024], and temporal localization [Huang et al., 2024]. VideoTree [Wang et al., 2024] and VideoAgent [Fan et al., 2024] caption keyframes before passing them to an LLM together with a question for answer generation, addressing the limitations of [Zhang et al., 2024a], which performs a similar methodology without keyframe selection on shorter videos. Recently, EgoBlind [Xiao et al., 2025], the first egocentric VideoQA dataset collected from blind individuals, showed that current VLMs perform poorly on assistive tasks. Similarly, Vid2Coach [Huh et al., 2025] highlighted the limitation of VLMs for blind assistance due to lack of finegrained analysis. In this project, we will also leverage LLMs, both in our VLM use, as well as in dialogue modeling.

In terms of long-form video understanding, the definition of long videos has evolved with the release of various datasets spanning seconds [Xiao et al., 2021], a few minutes [Grauman et al., 2022], 10-30 minutes [Argaw et al., 2024], or one hour [Islam et al., 2024]. Increased video lengths led to a range of works focusing on efficient temporal modeling strategies. A common technique to deal with longer videos is to use pre-extracted visual features [Han et al., 2023a]. For end-to-end learning with transformers, several works explored factorized

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spatio-temporal attention [Bain et al., 2021]. Others have looked at various ways to incorporate a memory mechanisms [Kahatapitiya et al., 2024], or captioning frames to exploit LLMs [Zhang et al., 2024a; Wang et al., 2024]. In this project, we will similarly leverage LLMs to effectively store long-term memory in text form.

Task-oriented dialogue (TOD). Task-oriented dialogues aim to successfully accomplish tasks by accessing external and reliable information while being fluent and natural. Traditional pipelines integrate specialized natural language processing components and optionally speech recognition and synthesis [Ultes and Rojas-Barahona et al, 2017; Cordier et al., 2020; Chen et al., 2018; Cordier et al., 2022]. Unfortunately, these approaches require separate specialized components, involving extensive engineering. Neural network architectures that built on top of pretrained language models provide more flexibility and potentially better fluency, but require large training corpora and are prone to hallucinations. LLMs [Ouyang et al., 2022] generalize from instructions or a small number of examples and promise fluent and natural replies. Recent approaches explore simple prompting [Hudeček and Dusek, 2023] or advancing agentic prompting [Elizabeth et al., 2025]. Nevertheless, results show that LLMs still underperform in task-oriented dialogue, reporting very low success rates [Hudeček and Dusek, 2023; Bonial et al., 2024; Elizabeth et al., 2025]. Typically, LLMs capabilities are enhanced by utilizing reinforcement learning (RL), by training reward models on distinct criteria (e.g. helpfulness, harmlessness and safety [Ouyang et al., 2022]), by combining deterministic rewards with reward models to enhance both reasoning and language abilities [Liu et al., 2024] or by incentivizing clarifications [Acikgoz and Oh, Jinoh et al., 2026]. In this project we will explore RL to improve VLM abilities in realistic or specific changing environments and to learn policies specialized for the VI assistance tasks.

Preliminary results. The navigation research from Shizhe Chen (Inria) has explored 3D mapping [Chabal et al., 2025; Li et al., 2025, 2026], various scene memories for visual navigation [Chen et al., 2021, 2023, 2022b], and data-centric approaches to improve the generalization in unseen environments [Guhur et al., 2021; Chen et al., 2022c]. While these methods achieve state-of-the-art performance, most existing navigation environments are simplified as discrete graphs, neglecting the challenges in continuous 3D spaces, and fail to incorporate open-vocabulary understanding essential for real-world applications. This project aims to enhance generalizability for open-vocabulary object navigation in continuous environments, while improving navigation efficiency.

A series of works from Gül Varol (ENPC) has built VLMs to generate audio description captions for movies for the VI audience [Han et al., 2023a,b, 2024; Xie et al., 2024]. The so-called AutoAD series have achieved automatic audio description (AD) prediction given long videos, either via training VLMs [Han et al., 2023a,b, 2024], or by developing a zero-shot pipeline in more recent work [Xie et al., 2024]. While the results are promising, the VIDI project will go beyond passively captioning videos, and explore dynamically analyzing the video content based on user inputs through a dialogue with the VI.

Previous work at Orange has focused on policy learning for task-oriented dialogue, such as budgeted reinforcement learning [Carrara et al., 2019], structured policies and imitation learning for multi-domain dialogues [Cordier et al., 2020, 2022, 2023]. We have also evaluated the capabilities of LLMs on TOD by using advanced prompting, reporting a low success rate [Elizabeth et al., 2025] and worked on the interpretability of visual models [Cafagna et al., 2023].

c Methodology and risk management

The project is organized into five Work Packages (WP). Figure 2 illustrates their relation and Figure 6 their timeline.

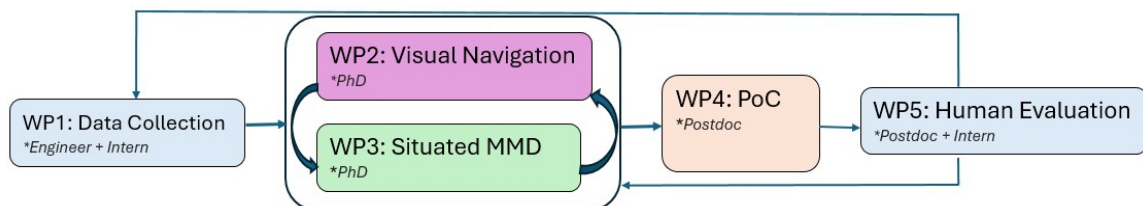


Figure 2. Work-package interactions and requested resources. The postdoctoral fellow of WP4 will also contribute to WP3 (detailed involvement is presented in Table 2).

Work Package 0: Project Management

Leader: L. Rojas-Barahona (Orange)

Participants: ENPC, Inria, IRIT

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This work package oversees the global coordination of the project (Task 0.1). Moreover, it includes the administrative and financial management task (Task 0.2), during which relations with funding agencies are handled. It also includes the ethics and data management plan (Task 0.3), which ensures compliance with regulations, as well as the dissemination and communication strategy (Task 0.4).

Task 0.1: Coordination

This task concerns the coordination of the research efforts by aligning methodologies and promoting collaborations between the consortium. It regards the accomplishment of the objectives of all the WPs and their milestones. It lasts for the entire duration of the project (*48months*). A Management Committee composed of WP leaders and the coordinator will be established to ensure efficient internal communication. At least one in person meeting per year is envisaged to provide feedback on the progress, risks and future directions. Complementarily, monthly virtual meetings will be held to ease the coordination among the work packages. Further exchanges and extended visits will be encouraged to support the collaborations between partners.

Task 0.2: Administration and finance

This task oversees the administrative and financial management of the project, as well as the interaction with the ANR. The project coordinator will verify with the Management Committee that resources are well invested and that deliverables are submitted on time. The consortium agreement (Deliverable D0.2) and the intellectual property considerations will be treated in this task as well as the final report (Deliverable D0.4).

Task 0.3: Ethics and data management plan

This task concerns the preparation of the dataset protocol for the Ethics Committee and the General Data Protection Regulation (GDPR) entities, particularly for the partners involved in the data collection (WP1) and the human evaluation (WP5). We advocate for open-source datasets and models with reproducibility in mind, in accordance with the GDPR protocol. The data management plan should be submitted after 6 and 24 months of the beginning of the project and at the end of the project (Deliverables D0.3).

Task 0.4: Dissemination

This task focuses on the communication strategy of the project's findings. A website will serve as a hub for publishing relevant information, such as internal and/or external events, project achievements and scientific publications. The results of the project will be shared with the research community through international conferences and journals in the areas of natural language processing, computer vision, and machine learning. Additionally, we will set up a biannual newsletter to inform participants about the project's progress. At the end of the project, we will organize a feedback workshop with the participants to present the project results and to explore future perspectives with them. Important milestones that we will share with the community include the high-quality dataset, the visual navigation and situated dialogue models as well as the evaluation outcomes.

Deliverables.

- D0.1: M3 - Website of the project
- D0.2: M6 - Consortium agreement
- D0.3: M6,M24,M48 - Data management plan (DMP)
- D0.4: M48 - Final Report

Risk management:

Risk 0.1: Delay in the recruitment of Postdoctoral and PhD students.

To mitigate this risk, we have opted for a 48-month project duration, which allows for a longer recruitment period to ensure that we find the best candidates for our demanding topic. Additionally, we will consider applications from our own master's and PhD students for the available positions.

Work Package 1: Data Collection

Leader: Antonio Serpa (IRIT)

Participants: IRIT (Tasks 1.1, 1.2 and 1.3); Orange (all tasks); ENPC and Inria Paris (Tasks 1.1 and 1.2).

In this work-package, we will focus on collecting dialogues between VI and their guides. After approval of the Research Ethics Committee (Task 1.1), we will define, test and validate some representative scenarios with a small group of VIP, caregivers and professionals (Task 1.2). After the validation of these scenarios, we will recruit 25 VIP and guides to acquire the dataset (Task 1.3). For safety reasons, we will limit the evaluation data to indoors, although the developed methods will be applicable to outdoors. The collection will take place

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in the semi-realistic environments of the “Maison Intelligente de Blagnac” (MIB⁵) and the Coffee Lounge (CL) in Orange Innovation at Lannion. Finally, the collected dialogues will be transcribed and further annotated semi-automatically with grounding and planning (Task 1.4).

Task 1.1: Ethical committee proposal and GDPR validation

Before the scenario definition, we will submit the dataset protocol to the Ethics Committee of the University of Toulouse and Orange. In parallel, we will ensure that our approach complies with the GDPR. We will address the feedback and concerns raised by the Ethics Committee whenever necessary. Although this task is closely related to the Task 0.3, data management plan, it concerns solely the organizations involved in the data collection process.

Task 1.2: Scenario definition

We decompose this task into three sub-tasks: needs assessment, scenario design and validation process.

Needs assessment. The needs analysis within the VIDI project will require a *participatory approach* involving both professional and family caregivers. The first step will be to *gather the needs and expectations of users*. To achieve this, *semi-structured interviews* will be conducted with caregivers and visually impaired individuals (about 4 or 5 participants) to identify the challenges they face in *indoor navigation* and *locating everyday objects*. *Focus groups* bringing together professional and family caregivers (occupational therapists, mobility instructors, personal care assistants, and relatives) will facilitate discussions on *existing practices* and *compensatory strategies* already in use. Additionally, *targeted questionnaires* may be distributed to refine the understanding of the expectations of different stakeholders. This phase of *qualitative and quantitative data collection* will rely on well-established methodologies in *ergonomics and social sciences*, such as those developed by [Bastien and Scapin, 1993] on *user needs analysis* and *human-computer interaction*.

Scenario design. The co-construction process of experimental scenarios also should adopt a participatory approach based on IRIT works [Zhao et al., 2022; Serpa et al., 2024], where all stakeholders, including VIP, professional and family caregivers, as well as researchers and engineers, are involved in the development of usage scenarios. This will begin with organizing co-design workshops bringing together these various actors to develop realistic scenarios, taking into account the specific needs of end users, such as obstacle detection, object recognition, and multimodal dialogue (between VIPs and caregivers) to facilitate navigation. For example, Figure 1 shows a concrete scenario that could be implemented and tested. These workshops will also help to better understand the practical constraints related to each situation and, ultimately, propose a solution tailored to the users' reality.

Validation process. Once the experimental scenarios have been co-constructed, they will be validated and refined through a series of user tests in the controlled environments of MIB and CL. A small panel of visually impaired individuals (2 to 3 participants) will carry out the different scenarios by interacting with a “Wizard Of Oz” experimental platform. This will allow detailed feedback to be gathered on their user experience. The VIPs will assess the complexity and feasibility of the scenarios in real-life contexts. In parallel, caregivers, will play a key role in observing and evaluating the impact of the solutions on the autonomy and comfort of the visually impaired individuals. The feedback from these tests will be used to iterate the scenarios and adjust the prototype, in order to optimize the proposed solution according to the users' real needs and to improve the usability and acceptability of the system.

Task 1.3: Recruitment and data collection

We consider in this task the three steps described below.

Participant recruitment. Participants will be recruited through two methods: a minority will be selected via researchers' personal networks, while the vast majority will be recruited following an announcement sent by email to the designated contacts of the joint IRIT/IJA laboratory “Cherchons pour voir”⁶, of which Bernard Oriola is a founding member and a member of the scientific council. We will also distribute this announcement to the representatives of the partner associations collaborating with IRIT, such as the disability center of the José Cabanis media library, UNADEV, AVH, Retina France, etc., who will ensure the internal dissemination of this call for participation in the study. The participant panel (approximately 25 individuals) will consist of visually impaired individuals with a visual acuity of less than 3/10, according to the WHO classification, and who are fluent in French. To consider the broader ecosystem of assistance, we will also include professional and/or family caregivers in this study.

Experimental setup. The data collection will take place in the MIB and CL. We will welcome participants

⁵<https://mi.iut-blagnac.fr/>

⁶<https://cherchonspourvoir.org/>

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Figure 3. Example of the semantic annotation.

individually, accompanied or not by their caregivers. We will prioritize the guide to be the individual's caregiver if they give consent, otherwise one of the IRIT/Orange members will act as the guide. After reading and approving the consent form (a Braille copy will be made available), we will have the participants complete a socio-demographic questionnaire. Subsequently, we will equip the participants with wearable cameras that have integrated audio (likely Meta ARIA camera glasses or a smartphone). We hypothesize that camera glasses are the best option; however, some VIPs may have difficulty using them due to limited head mobility. Therefore, if necessary, we will offer smartphones as an alternative device. After being equipped, we will position them within the platforms (MIB and CL) to begin executing various scenarios (see Task 1.2). The participants will carry out several scenarios (about 10min each) of varying complexity, with a 10-minute break between each scenario. Regarding the technical requirements, in this step we need to interface with the assistive glasses or other alternative devices, to receive the audio and visual inputs and to correctly record the interactions. We will record the conversation between the camera wearer VI and the sighted guide, who will be holding a screen to follow what the devices can *capture*. The audio and video stream will be recorded on a dedicated PC in order to transcribe the data into text later for the purpose of creating the dataset (see Task 1.4). At the end of the session, a debriefing based on a semi-structured interview will be conducted to gather information about the user experience.

Safety measures. Moving a visually impaired person through an unfamiliar environment filled with obstacles can be risky. Therefore, to minimize these risks as much as possible, a human operator will be positioned near the participant to intervene at any moment in case of major risks. This will be complementary to the navigation and obstacle-detection assistance provided to the participant by the experimenter/caregiver through the audio and video stream feed from the camera glasses and relayed to the observation room.

Task 1.4: Data annotation and completion

The dialogues collected from the audio will be transcribed using automatic speech recognition (ASR) techniques. Specifically, we will employ WhisperX [Bain et al., 2023] with the diarization module, which provides timestamps for each utterance, together with the speaker identity in case of multiple speakers.

We will further annotate the dialogues manually, focusing on grounding (do speakers agree in their mutual understanding?) and planning (was the task effectively accomplished?). To achieve this, the semantic representation must clearly define the VI assistance tasks, allowing for an effective evaluation of conversations later in (WP3).

We will build an evaluation dataset upon the collected high-quality data. This step, carried out by Orange, is of great importance to formalize the VI assistance tasks, which might involve obstacle detection, object recognition, scene description and/or navigation guidance. We will annotate the dialogue interactions with a semantic representation at turn-level that contains dialogue acts (or communicative acts) such as: Request, Inform, Offer, Command, Accept, Reject, etc. We will also consider grounding acts such as: Repeat, Explain, Clarify, Self-correction, Self-repetition, Repair, Done; and propositional content such as: Move, Turn, Enter, Exit, Approach, Go. We will also provide global dialogue annotations regarding the goal of the VI and the success rate. The annotation will also include coreference resolution for referencing objects that were previously discussed or are present in the scenes.

To give a specific example, consider the first two utterances in Figure 1 of the VI and the guide respectively, which might be annotated as in Figure 3. However, it is worth noting that one major step is to clearly define the annotation scheme (Deliverable D1.4), which will be provided to outsourcing annotators. The goal is not to have a complex hierarchical representations as proposed in [Bonial et al., 2024]. In contrast, we opt for simpler representations as in SIMMC2.1 [Kottur and Moon, 2023]. We will consider augmenting the set of annotated data semi-automatically whenever the manually annotated data is insufficient.

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Deliverables:

- D1.1: M4 - Ethical and GDPR proposal
- D1.2: M7 - Scenarios for VI assistance
- D1.3: M12 - Initial dataset
- D1.4: M14 - Multimodal annotation guidelines
- D1.5: M18 - Dataset with annotations

Risk management:

Risk 1.1: Delay on the ethical approval process

The validation process with an ethics committee can sometimes take time if it does not comply with the ethical and data protection guidelines recommended in the field of research. Antonio Serpa is one of the founding members and a current member of the CER (Research Ethics Committee of the University of Toulouse). He is able to self-assess the files we submit to ensure they meet the committee's expectations. The risk of delay, while controlled, remains present but only marginally.

As a reminder, the released data will be used for the evaluation of the models. WP2 and WP3 can start working under simulation and/or with synthetic data, created from existing VIP data such as VizWiz [Chen et al., 2022a] and EgoBlind [Xiao et al., 2025].

Risk 1.2: Delay/difficulties in recruiting VIP

Recruiting visually impaired participants can sometimes be lengthy and complex. That is why, in order to conduct research in the field of digital assistance tools for visually impaired individuals and to simplify the volunteer recruitment process, IRIT has established a joint laboratory with the "Institut des Jeunes Aveugles" and has also formed partnerships with a range of associations and institutions working in favor of VIP (see Task 1.3).

Despite these efforts, the recruitment process for participants can sometimes take longer than initially expected. To mitigate this risk we will consider initial data collection with sighted participants simulating visual impairment while awaiting approval.

Work Package 2: Training Generalizable Vision-and-Language Navigation (VLN) Models

Leader: S. Chen (Inria)

Participants: Inria in all tasks; ENPC in Task 2.2; Orange in Task 2.2.

This work package focuses on developing navigation systems capable of guiding VI users to explore their surroundings and locate objects using language instructions. To achieve this, we propose a modular framework consisting of three core modules, as illustrated in Figure 4a:

- Semantic mapping: continuously construct and maintain a scene memory with semantic information based on sequential visual observations from egocentric cameras.
- High-level goal prediction: given language instructions from VI users, predict a target location in the constructed map as a high-level navigation plan.
- Low-level path generation: translate the high-level goal into obstacle-free path trajectories to guide VI users toward the target.

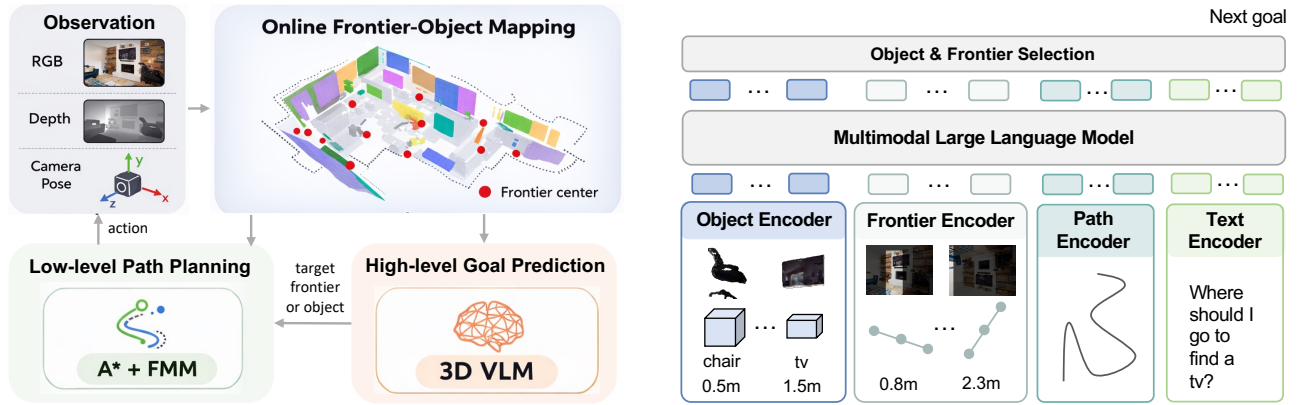
Therefore, this work package is divided into three tasks. Task 2.1 aims to investigate optimal semantic map representations and to develop efficient methods for real-time construction of such maps from RGB-D observations. Task 2.2 will explore multimodal language models to enable generalizable VLN in unseen environments, efficiently integrating scene memory, user instructions, and path histories to predict the high-level navigation goal. Finally, in Task 2.3, we will develop efficient low-level navigation policies to adapt in dynamic environments and to provide timely assistance for VI users.

Task 2.1: Online 3D semantic mapping

This task aims to construct a geometrically and semantically rich scene memory from streaming RGB-D camera data. To balance memory efficiency and semantic richness, we will develop a hybrid map representation, consisting of a sparse 3D point cloud of the scene, 2D obstacle and exploration maps, and an object-centric scene graph with latent semantic representation per object.

Building such hybrid maps requires localization, 3D reconstruction, semantic segmentation, and map accumulation. We first will utilize well-developed simultaneous localization and mapping (SLAM) systems [Mur-Artal and Tardós, 2017] to estimate camera poses from sequential RGB-D observations. Given an accurate camera pose at each step, we can project the RGB-D observation into a unified 3D point cloud, and merge the new point cloud with the previously stored point cloud. The merged point cloud will be further downsampled to maintain memory efficiency. From this scene point cloud, we can efficiently derive 2D obstacle and exploration

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(a) Overall framework of the vision-and-language navigation system, consisting of three key components: online 3D semantic mapping, high-level goal prediction and low-level path planning.

(b) The architecture of the high-level goal prediction model. The model first encodes the constructed semantic map, path history and language instruction, and then uses MLLMs to predict the next goal.

Figure 4. Overview of the proposed modular vision-and-language navigation framework.

maps. For constructing the object-centric scene graph, object segmentation is necessary. However, directly applying a pretrained 3D segmentation model to process the sparse scene point cloud is inefficient and can suffer from lacking fine-grained shape and texture details. In addition, existing pretrained 3D segmentation models are less satisfactory due to data scarcity compared to more mature 2D segmentation models. To address these issues, we will leverage the state-of-the-art 2D segmentation model SAM [Ravi et al., 2024] for object segmentation. We will extend its capabilities to continuously track new objects over time. Furthermore, we will train specialized 3D modules to lift features from 2D to 3D and dynamically group new object features with stored object memories, creating coherent object-centric representations in 3D space. This approach will enable accurate and efficient scene memory updates during navigation.

We will evaluate the online 3D semantic mapping approach on standard benchmarks including ScanNet [Dai et al., 2017], MP3D [Chang et al., 2017] and HM3D [Ramakrishnan et al., 2021], measuring performance on 3D reconstruction and segmentation as well as computational efficiency.

Task 2.2: Generalizable navigation goal prediction model

This task aims to develop a generalizable vision-and-language navigation model capable of predicting high-level goals in diverse environments.

We will build our model on top of large vision-language models (VLMs) [Chen et al., 2024b] to improve generalization, reasoning and planning capabilities. VLMs can enable more generalizable multimodal grounding between texts and visual objects, and improve reasoning of object affordances (e.g., recognizing that a water bottle but not biscuits will fulfill the goal of finding something to drink). Moreover, they can provide commonsense knowledge for exploration (e.g., knowing that the water bottle is likely stored in a fridge and prioritizing exploration to a kitchen). However, VLMs typically consist of an image encoder and a large language model, which are not optimal for navigation tasks. Hence, we will extend VLMs with additional encoders to integrate our constructed scene memory and path history, as illustrated in Figure 4b.

To support model training, we will construct a comprehensive dataset by combining existing annotated navigation datasets such as R2R, REVERIE, SOON, HM3D, and MP3D. Additionally, we will generate more navigation training data, leveraging our prior experience in data construction [Guhur et al., 2021; Chen et al., 2022c]. While our prior works rely on template-based methods to automatically generate high-level and step-by-step instructions, we will explore using LLMs and VLMs to improve the quality of generated instructions. For training, we will explore parameter-efficient fine-tuning methods such as LORA [Hu et al., 2022] to preserve the model's general knowledge while optimizing for navigation-specific reasoning. Different training tasks will be investigated including next navigation goal prediction, and general visual question answering to enhance models' understanding and reasoning capabilities. For evaluation, we will benchmark performance across multiple VLN datasets especially GoatBench [Khanna et al., 2024] which closely simulates real-world navigation scenarios for VI users. Key metrics will include navigation success rate and success rate penalized by path length which further considers navigation efficiency.

Task 2.3: Efficient low-level path planning policies

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This task aims to develop an efficient low-level path planning policy to reach a navigation goal specified by coordinates. Instead of relying on analytical path-planning solutions, we propose a learning-based policy to better adapt to dynamic environments with moving elements such as pedestrians and pets. Unlike traditional methods, a learning-based approach can inherently leverage past observations to estimate trajectories of moving elements and then predict appropriate actions, improving coherent guidance for VI users. To enhance efficiency, we will design lightweight navigation policies that process the constructed map, instant visual observations and the target goal. We will explore architectures such as RNNs and transformers, following our previous work [Chen et al., 2023], and train the model using a combination of behavior cloning and reinforcement learning. Once the navigation policy predicts the optimal trajectory, the model can communicate it to the VI user through auditory feedback.

Risk management:

Risk 2.1: Computational constraints limiting real-time feedback for VI users.

To address this risk, we will place the computationally intensive goal prediction model to cloud servers, while only putting the lightweight low-level policy on device to ensure responsive feedback.

Risk 2.2: The simulation-to-real gap.

Although the simulated environments used for model development are from real-world scans, there are still noticeable visual differences compared to real observations. To mitigate the gap, we will explore domain randomization approaches and combine real-world images in training.

Deliverables and success indicators. For each task, we will summarize our findings in research reports and open-source the associated software including codes and models. The publication of these works in top-tier computer vision and robotics conferences will be one of the key success indicators for this work package. In addition, performance improvements on standard benchmarks will serve as another success indicator.

- D2.1: M20 - Publication and software on online 3D semantic mapping
- D2.2: M33 - Publication and software on high-level navigation goal prediction model
- D2.3: M42 - Publication and software on efficient low-level path planning policy

Work Package 3: Situated Multimodal Dialogue

Leader: G. Varol (ENPC)

Participants: Orange (Task 3.1), ENPC (Tasks 3.2, 3.3, 3.4).

We structure this work package into tasks that focus on dialogue modeling (expertise of Orange) and video modeling (expertise of ENPC) separately, but these tasks will progress with synergy and benefit from each other. We divide the work into four tasks: we explore reinforcement learning for dialogue policy (Task 3.1), modeling long-term video context (Task 3.2), moment detection for turn taking and environment changes (Task 3.3), and self-supervised learning for egocentric video representations (Task 3.4).

In summary, on the dialogue side, we will first benchmark the capabilities of existing VLMs on VI assistance tasks, and further improve them with reinforcement learning (RL). On the visual modality side, we will develop capabilities to model long-form videos by storing key visual elements in the form of text, which can later be ingested by LLMs to incorporate long-term context (building on our recent work [Ventura et al., 2025]). Moreover, we will work on identifying changes in the environment that are worth describing to the user. We will also develop visual representations tailored for egocentric videos using self-supervised learning on external data sources that capture long daily activities from wearable cameras.

Task 3.1: Reinforcement learning for dialogue policy

We will start by evaluating the zero-shot performance of state-of-the-art VLMs [Li et al., 2024] on grounding and planning capabilities for VI assistance on the evaluation dataset collected in WP1. We will meticulously conduct error analysis and VLM interpretability (as we explored in preliminary work [Cafagna et al., 2023]). This step will enable us to devise ways to enhance VLMs. To this end, we will first identify the most recent and efficient open-source visual language models, and we will evaluate their performance for situated multimodal dialogue and VI assistance. We will first conduct quantitative and qualitative evaluations of these models using available multimodal situated dialogue datasets that have been utilized to evaluate grounding and planning. Examples of these datasets are SIMMC 2.1 [Wu and Kottur, Satwik et al., 2023], utilized in the Dialogue System Technology Challenge DSTC-11 [Kottur and Moon, 2023], the VizWiz challenge [Chen et al., 2022a], a benchmark of Visual Question Answering for VIP and more recently EgoBlind Xiao et al. [2025]. Then, we will go further in the analysis of VI assistance tasks, according to the scenarios and annotations defined in the WP1 (Tasks 1.1 and 1.4). This step will help us to identify frequent VLM errors and to devise ways to

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solve them, either through advanced prompting [Yao et al., 2023] or through reinforcement learning.

As suggested by recent work, VLMs are far from perfect when performing planning in task-oriented dialogue [Elizabeth et al., 2025; Acikgoz and Oh, Jinoh et al., 2026; Xiao et al., 2025]. To improve this, we will next explore the integration of reward models during training. We can either explore RL from human feedback (RLHF) to improve the abilities of pre-trained VLMs in realistic (or specific) environments, or learn policies specialized for the VI assistance tasks. We will compare VLMs with classical approaches such as (hierarchical) RL with imitation learning that we studied in [Cordier et al., 2022]. Since it is difficult to reproduce the exact dialogue from a given dataset, we simulate dialogues with simulated users and environments, as in well-established task-oriented dialogue benchmarks⁷, both to train RL algorithms and to evaluate model performance (e.g. success rate, task completion, user satisfaction, etc). Typically, to simulate users, we create realistic scenarios and user goals based on real data ([WP1]). However, incorporating the visual environment (e.g., AI Habitat⁸) and its changes poses a significant challenge, necessitating close collaboration with WP2. We will release the simulated data, models, and benchmarks to the research community. A postdoctoral fellow will be hired to work on this task.

Task 3.2: Modeling long-term video context

Current VLMs are not designed for feeding in more than a handful of RGB video frames at a time [Bai et al., 2025], which limits their capability to incorporate more than a few seconds as visual input. On the other hand, VI assistance operates on continuous video stream, typically requiring to remember the past guidance experience, not to repeat the same information, to be able to refer to previously seen visual elements, or to detect changes (see T3.3). Simply feeding all the frames to a VLM is impractical due to the heavy computational cost of processing videos. We will address this by keeping long-term context in the form of text. Such memory models are studied in [Kahatapitiya et al., 2024] for video QA. In contrast, here, our memory will consist of not only the visual context, but also the dialogue context. Specifically, we will use image captioning models to convert the past RGB frames into text, which are efficient to store and ready to be processed by LLMs. We will interleave this information with the previous dialogue interaction. Additionally, we will add the timestamp information for both captions and dialogues. We will have modules to update this memory (with write/delete functionalities as in [Kahatapitiya et al., 2024]) as the dialogue progresses. This textual memory can be appended to the prompt when generating the next response with a VLM besides the current video frames (see Fig. 5). This will be similar in spirit to our recent work [Ventura et al., 2025], where we leverage LLM finetuning to perform efficient long-video segmentation into semantic units, by providing frame-level captions and transcribed speech. In this project, we will initially test zero-shot techniques; however, if we find the need for finetuning the VLM to adapt to the new prompt structure, we will adopt LoRA [Hu et al., 2022] fine-tuning strategy potentially on the recent VSI-Bench [Yang et al., 2024] dataset, which provides egocentric videos equipped with QA pairs to evaluate a number of visual-spatial intelligence tasks about 3D configuration of a scene such as absolute and relative distances of objects, and route planning.

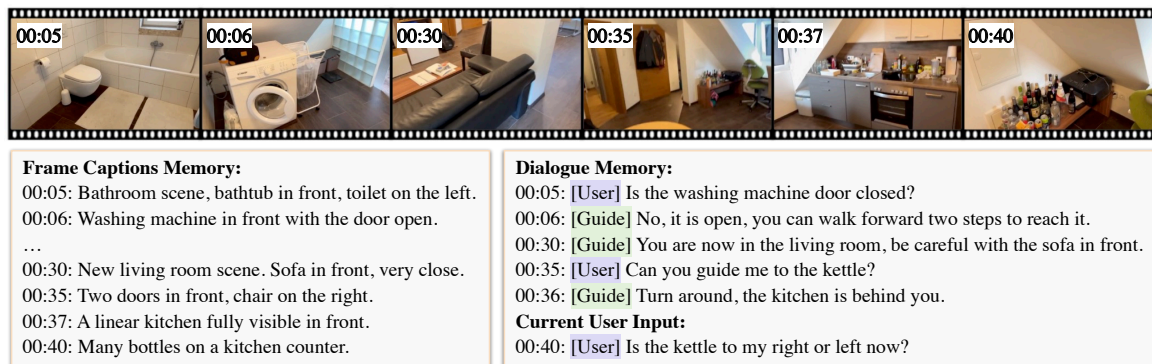


Figure 5. Illustration of long-term context, where the captions describe previous video frames, and the dialogue between the user and the guide, both stored in text form with their timestamps. The example image is taken from the VSI-Bench dataset [Yang et al., 2024], and the dialogue is complemented artificially.

Task 3.3: Moment detection for turn taking upon environment changes

An important visual task is to detect the moments when the machine should intervene given the egocentric video and the dialogue context. We will develop multiple functionalities such as detecting (a) new people

⁷<https://github.com/ConvLab/ConvLab-3>

⁸<https://aihabitat.org/>

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in the scene, (b) obstacles such as stairs and obstructing objects, (c) changes in the environment such as new room, and (d) anomalies potentially indicating a dangerous situation (e.g., people running). We will request the user to define their preference upfront whether they wish to be interrupted with such change detections. We will use a combination of face detection [Deng et al., 2020] for new people and VQA [?] for obstacle and new environment detection by formulating a series of questions. When a change detection is triggered, the machine will initiate a new turn without necessarily being prompted by the user. We will evaluate the moment detection performance on the data collected from WP1.

Task 3.4: Self-supervised learning for egocentric video representations

While we will employ powerful VLMs, which are considered foundation models in the sense they successfully generalize across visual domains, we are aware of their lowered performance when applied to fisheye egocentric images [Grauman et al., 2022]. To improve visual features, we will focus on self-supervised learning to train visual encoders specialized on egocentric videos captured from wearable cameras. We will leverage external large-scale data sources such as Ego4D [Grauman et al., 2022], and again use LoRA [Hu et al., 2022] to adapt the vision backbone of a pre-trained VLM with a contrastive objective as in DINOv2 [Oquab et al., 2024], while keeping the text generation objective using the caption annotations where available.

Risk management. We are aware of the challenges associated to the latency in real-time interactions. To mitigate this risk, the dialogue models can be stored on a local GPU server to lighten the communication burden with cloud computing. Frugal models will also be taken into consideration.

Deliverables and success indicators. Similar to WP2, we will summarize our findings in research reports. We will aim to publish them in top-tier computer vision, machine learning, and natural language processing venues, which will be the success indicators for this WP, as well as the performance on benchmarks. For each publication, we will make the associated software, including the code and trained models, publicly available for research.

- D3.1: M30 - Publication and software on situated-multimodal dialogue models
- D3.2: M18 - Publication and software on long-term video context
- D3.3: M30 - Publication and software on moment detection
- D3.4: M42 - Publication and software on self-supervised learning

Work Package 4: Proof-of-concept

Leader: L. Rojas-Barahona (Orange)

Participants: Orange (Tasks 4.1, 4.2 and 4.3).

The situated MMD system for VIP assistance tasks will be implemented by utilizing the model(s) built in WP2 and WP3. An experimented research engineer from Orange will work on this proof-of-concept (PoC) in collaboration with a Postdoctoral fellow and contributors of other WPs (e.g. ergonomists). The goal will be to deploy a prototype with the smart glasses or alternative devices, potentially using edge computing to minimize the latency. **This prototype will be crucial for the technological transfer among the operative entities at Orange.** We divide the work into the following tasks.

Task 4.1: Hardware requirements: devices and high-performance computing

Following the data collection from Wizard of Oz experiments (WP1 task 1.3), the prototype also needs to interface with the assistive glasses or other alternative devices, to receive the audio and visual inputs and to correctly record the interactions. We will opt for a cascade approach that requires automatic speech recognition (ASR) to transform the audio into input texts and text-to-speech (TTS) to transform text responses back to audio. A technical analysis of the high-performance computing (HPC) requirements, as well as latency issues will be also carried out in this task. In addition, we need to assess the possibility of using edge solutions.

Task 4.2: Software integration: visual and dialogue model

In this step we will work closely with WP2 and WP3 to define efficient ways to integrate the models, considering hosting them for inference in GPU high-performance computing (HPC). We will probably face optimization issues, and we might need to further parallelize the GPU usage.

Task 4.3: Packaging and deployment

We will build a prototype, which consists in a pipeline that includes the following components: device interface, ASR, navigation and dialogue models and TTS. Basically, it will receive the video-audio inputs, transcribe the audio, utilizes the models for navigation and multimodal situated dialogue and synthesizes the output. We will involve ergonomists to ensure that the PoC is well suited for VIPs.

Deliverables:

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D4.1: M44 - Prototype.

Success indicators: demonstration publication.

Work Package 5: Human Evaluation

Leader: Antonio Serpa (IRIT)

Participants: IRIT (all tasks); Orange (Tasks 5.1 and 5.3), ENPC and Inria (Tasks 5.3).

Besides the automatic evaluations conducted in each WP, we will target VI user evaluations. Upon approval of Research Ethics Committee of the University of Toulouse, this work package will implement user experience evaluation methods and tools to validate the utility and usability of the PoC for VIP. This WP will contribute to produce inclusive evaluation benchmarks for dialogue in VI assistance and the impact of AI on interaction processes. We divide the work into the following tasks.

Task 5.1: Ethical and methodological validation

Before the evaluation, we will submit protocol to the Ethics Committee of the University of Toulouse. In parallel, we will ensure that our approach complies with GDPR. If needed, addressing feedback and concerns raised by the Ethics Committee.

Task 5.2: Protocol design, recruitment and experiment implementation

We will conduct both a quantitative evaluation by selecting and/or adapt quantitative metrics (error rate, task completion time, failure rate, etc.) and a qualitative evaluation by designing qualitative evaluation tools (questionnaires, verbatim responses, etc.) with a sample of approximately 15 participants. The evaluation will be based on the scenarios defined in WP1 and will compare two experimental conditions: a control condition with a human guide and without using the PoC; and a condition using the PoC without the human guide.

To address the challenge of recruiting visually impaired participants and to limit inter-individual variability, we will adopt a within-subjects approach. Each participant will complete the scenarios under both experimental conditions. To minimize learning biases, the order of scenario execution and condition exposure will be counterbalanced across participants.

Task 5.3: Data collection and analysis

In this subtask we will collect quantitative and qualitative data under semi realistic conditions (in the MIB platform). We will collect quantitative data such as the task and sub-tasks duration, completion rate (success or failure), number of dialogue turns relative to the interaction context, task complexity, and user profile. For the analysis of these data, we will use the R environment to perform ANOVA-type statistical analyses. If the population heterogeneity is too high and does not allow for meaningful statistical results, we will conduct case studies.

Furthermore, we will collect qualitative data such as sociodemographic questionnaires, audio recordings, meCUE, and audio for the final semi-structured interview. The models and annotation tools used in WP 1.4 will be used for annotating and analyzing navigation, recognition and object localization dialogues. Based on these annotations, we will apply statistical occurrence methods to the defined indicators to assess the level of satisfaction and acceptability of the PoC. These results will be cross-referenced with the metrics extracted from the meCUE questionnaire.

Deliverables:

- D5.1: M33 - Methodology and evaluation protocol
- D5.2: M48 - Results analysis

Success indicators: Scientific publication.

Risk management: The risks are the same as in WP1.

II Organization and implementation of the project

a Scientific coordinator and consortium

The consortium is composed of researchers with experience in natural language processing and dialogue (Lina Rojas-Barahona, Morgan Veyret and Tanguy Urvoy from Orange); video understanding (Gül Varol from LIGM, École des Ponts ParisTech); vision-and-language navigation (Shizhe Chen from Inria), as well as teams that work on enhancing the user experience for assistive technologies and responsible AI (Christele Tarnec and Valerie Botherel from Orange and Antonio Serpa and Bernard Oriola from Institut de Recherche en Informatique de Toulouse, IRIT). On the computer vision expertise, the two partners, "Imagine" (LIGM)

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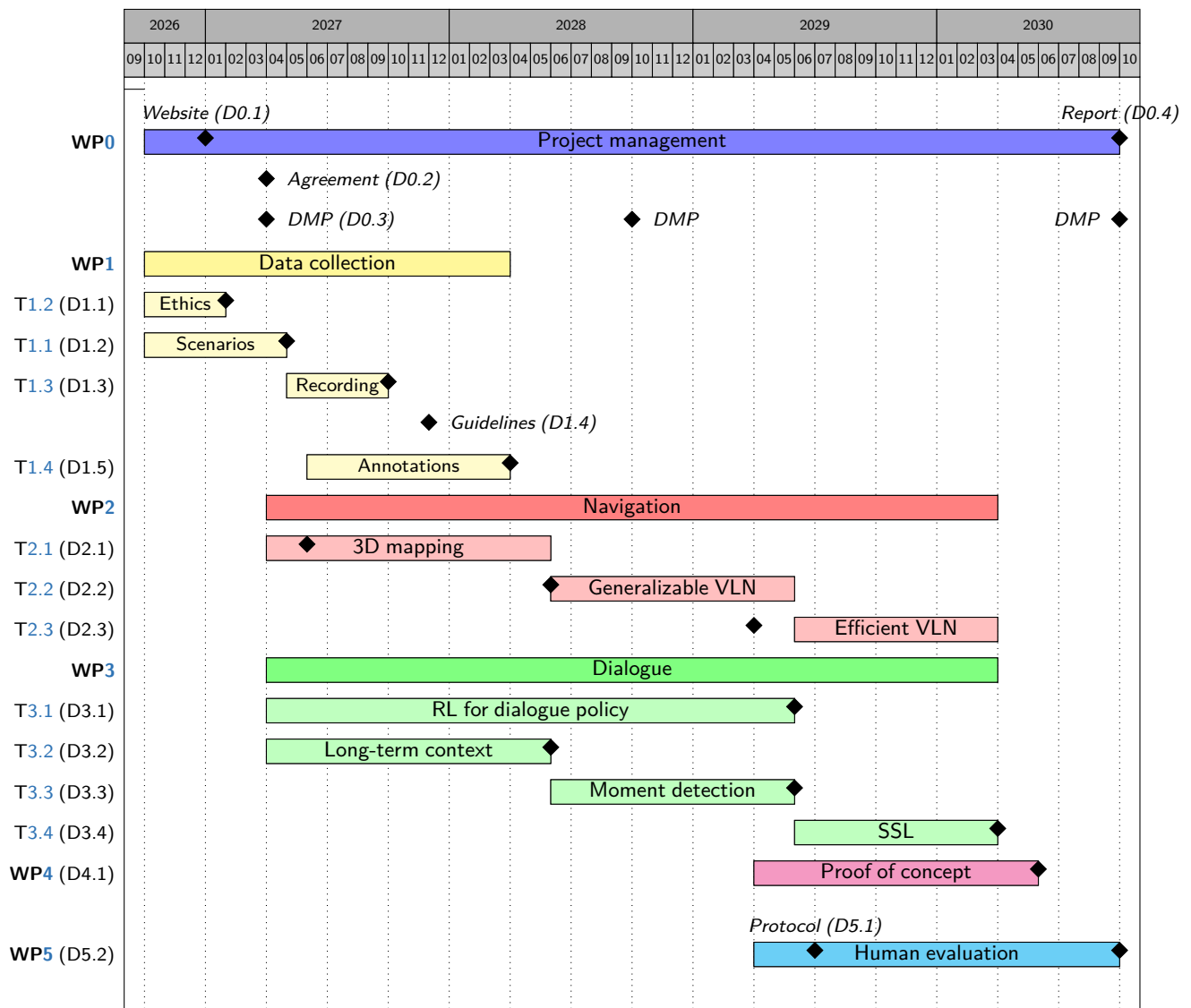


Figure 6. Gantt chart presenting the chronology of the WPs, associated tasks, and deliverables.

and “Willow” (Inria) teams, are internationally recognized in this domain. All members of the project already know each other and bring complementary expertise.

Coordinator: The scientific coordinator, HDR Lina Rojas-Barahona, program chair of the [EMNLP Industry Track](#) in 2025, is a senior researcher at Orange Research with extensive experience in Dialogue. She contributed to several collaborative research projects in Europe and in the UK and has [impactful publications](#). She was head of the 5-year industry project DIANA (Dialogue in Natural Language) with a budget of 800K € per year, involving not only contributors within Orange but also collaborations with academics through research contracts, PhD co-supervisions, and two Marie Skłodowska-Curie European projects: [COBRA](#) and [NL4XAI](#); yielding international scientific publications (e.g., [Cordier et al., 2020, 2022, 2023; Cafagna et al., 2023; Kejriwal et al., 2023]), as well as datasets and models released in Open-Source (e.g. CoQAR[Brabant et al., 2022], KGConv [Brabant et al., 2024]). She also contributed to the PyDial framework [Utes and Rojas-Barahona et al, 2017] and to the first neural dialogue pipeline [Wen et al., 2017] at the University of Cambridge, where she worked with Professor Steve Young and his team on statistical spoken dialogue systems.

Partner 1 Orange: (Lina Rojas-Barahona, Tanguy Urvoy, Morgan Veyret, Christele Tarnec and Valerie Botherel), before France Telecom, has long contributed to the research on dialogue⁹. Besides the project coordinator, Tanguy Urvoy is a senior researcher who has worked on theoretical aspects of reinforcement learning. He now works on deep reinforcement learning and generative models [Carrara et al., 2019; Fabre et al., 2020; Cordier et al., 2020, 2022, 2023]. A personal motivation for his involvement in this project is that his father is visually impaired. Morgan Veyret is an experienced research engineer with a PhD, who has worked on both applied research and deployment of tools and systems following best practices [Veyret et al.,

⁹Partner of the European project CLASSIC <https://classic-project.org/>

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2024; Elizabeth et al., 2025]. Orange is also engaged in responsible AI and accessibility. Christèle Tarnec, is a research engineer who works on this subject and Valérie Botherel is an ergonomist who specializes in accessibility and has experience working with individuals with disabilities.

Partner 2 ENPC: Gül Varol has strong expertise in large video-language models [Bain et al., 2021; Ventura et al., 2024, 2025], and she has already prior work applicable to assistive technologies (movie audio description generation for VI [Han et al., 2023a] and sign language translation [Varol et al., 2021; Jang et al., 2025]).

Partner 3 Inria: Shizhe Chen has remarkable experience in vision-language navigation [Chen et al., 2021, 2022b, 2023; Guhur et al., 2021; Chen et al., 2022c]. The proposed model [Chen et al., 2022b] has achieved the first place in the ICCV 2021 navigation challenges. She also has experience on 3D reconstruction [Chabal et al., 2025; Li et al., 2025, 2026] which is crucial for maintaining scene memory for navigation tasks. She is planning to co-organize the Embodied Agent and Dialogue workshop at ECCV 2026 with a challenge on dialogue-based navigation [Han et al., 2025].

Partner 4 IRIT: Antonio Serpa and Bernard Oriola will bring their expertise in user centered design and experience, especially in the domain of assistive technologies [Kammoun et al., 2012]. They collaborate with associations for VI [Serpa et al., 2024] and will be particularly involved in data collection and human evaluation. Of great importance, the contributors of IRIT are themselves visually impaired.

External collaborations: Each of the partners will have ongoing collaborations during the course of this project. For example, G. Varol will continue projects on movie audio description with Andrew Zisserman (University of Oxford), and the PhD co-supervision with Cordelia Schmid (Google) on 3D understanding from egocentric videos. Additionally, Meta has already provided to ENPC Aria glasses to be used for this project. S. Chen has been co-supervising a PhD student with Vincent Lepetit (ENPC) on active 3D mapping, a PhD student with Josef Sivic (Czech Technical University in Prague) on improving trustworthiness of large vision-language models. L. Rojas co-supervises with Professor Magalie Ochs of the University of Aix-Marseille a PhD student on the evaluation of conversations and a research collaboration started in 2025 with the same university on the same topic. L. Rojas and T. Urvoy co-supervised with professor Elisa Fromont from the IRISA a PhD thesis on tabular data generation. Pratik Gajane a professor chair at the University of Orléans and former student of T. Urvoy, will co-supervise with L. Rojas a PhD thesis on Multi-agent RL for dialogue, in line with VIDI dialogue research.

Name of the researcher	Person month	Funding source	Project title	Coordinator	Period
G. Varol	7	Hi! Paris Cluster 2030	TRAINER	G. Varol	2025–2028
	8	Korean Agency	Korean AI Hub	Kee-Eung Kim	2025–2028
	12	ANR-23-PEIA-0008	SHARP	LIP, ENS Lyon	2023–2028
	34	ANR-21-CE23-0003	CorVis	G. Varol	2022–2027
A. Serpa	10	AAP TIRIS BNI	LODYSEI	A. Serpa	2025-2027
	8	MIDOC	MaaS: Mobiliy as a Service	Valérie Camps	2022-2026
B. Oriola	5	IRESP	CMR-DV	Caroline Pigeon	2024-2028
S. Chen	34	ANR-25-CE23-7777	3D-GEM	S. Chen	2025-2029

Table 1. Implication of the scientific coordinator and partner’s scientific leader in on-going project(s).

	Orange	ENPC	Inria	IRIT	Total
WP0	3	1.5	1.5	1.5	7.5
WP1	20	1	1.5	39	58.5
WP2	6	1.5	42	-	49.5
WP3	25.5	38	-	-	63.5
WP4	33	-	-	-	33
WP5	6	-	-	37.5	40.5
	93.5	42	45	72	252.5

Table 2. Involvement of each partner in the different work -packages (in person-month).

b Implemented and requested resources to reach the objectives

We summarize, in Table 3, the breakdown of the requested sources.

Partner Orange

Orange reports below and in Table 3 the requested funding, which is 30% of the total expenses.

• **Staff expenses:** We will hire one postdoctoral fellow, who will work full time for 24 months on the dialogue policy (WP3) and the prototype (WP4) (189k€). We also include 85k€ for permanent and administrative staff.

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• **Instruments and material costs:** 2 wearable devices (1k€each) and one GPU server will be purchased for debugging locally the models, and for deploy and test the prototype (17k€).

• **Outsourcing / subcontracting:** we will hire qualified annotators to complete the semantic annotations (Task 1.4), the estimated cost is 30k€.

• **Overheads costs:** we estimate (6k€) for project meetings, involving all participants; 2 European conferences for at least two participants (10k€in total) and 1 conference outside Europe (4k€). We also considered a compensation of 3k€to VIP involved in the data collection and the evaluation tasks (WP1 and WP5).

Partner ENPC

• **Staff expenses:** One PhD student will be hired mainly for WP3 (135k€).

• **Instruments and material costs:** One GPU server will be purchased for debugging locally the VLMs, and potentially for testing the demo (15k€).

• **Overheads costs:** The travel costs will cover the international conference participation of the PhD student to present the three expected publications (1 per year), and the national trips to partners (15k€).

Partner Inria

• **Staff expenses:** One PhD student will be hired for WP2 (126k€).

• **Instruments and material costs:** one GPU laptop will be purchased for local development (3k€).

• **Overheads costs:** the travel costs will cover the PhD student's and PI's participation in local and international conferences and workshops for three years, requiring a total of 24k€, calculated as 4k€ per trip × 2 people × 3 years. We anticipate attending at least one international conference per year. Each trip costs approximately 4k€ to cover expenses such as registration fees, flights, accommodation and other travel-related costs.

Partner IRIT

• **Staff expenses:** One postdoc (18 months, 74 547€) in cognitive science will work on ethics approval tasks and the evaluations as well as on the recruitment of participants, testing, and analysis of the results (WP1 and WP5). One research engineer (12 months, 61 306€) will work on designing the scenario and creating the dataset (WP1). Two MS students (8.4k€) will assist the engineer and the postdoc.

• **Instruments and material costs :** 2 Smartglasses device and one smartphone/wearable computer (2k€), 2 computers for the postdoc and the engineer (2k€), as well as a high-performance computational unit (9k€).

• **Outsourcing / subcontracting:** : The rental of "Maison intelligente de Blagnac" platform will cost 15k€ (43 days * 350€). We also estimate a cost of 2k€ for publication fees.

• **Administrative and structure:** 26k€ for the administrative management and structure costs.

• **Overheads costs:** Participation in top international conferences (e.g., CHI, ASSETS, ICCHP. 3.5k€ per mission), as well as national conferences (1.5k€ per mission). We also estimate six additional national missions for the project-related activities (1k€ per mission) and 1k€ for a workshop organization.

	Orange	ENPC	Inria	IRIT
Staff expenses	274 015.81€	135 000€	126 000€	144 252.49€
Instruments and material costs	17 000€	15 000€	3 000€	13 000€
Building and ground costs	-	-	-	-
Outsourcing / subcontracting	30 000€	-	-	20 000€
Overheads costs	23 000€	15 000€	24 000€	19 000€
Admin. & structure costs	-	22 275€	20 655€	26 494.09€
Sub-total	344 015.81€	187 275€	173 655€	222 746.58€
Requested funding	927 692.39€			

Table 3. Requested means by item of expenditure and by partner.

III Impact and benefits of the project

a Social impact

The core of the project concerns the accessibility and inclusion of visually impaired individuals through technology that facilitates their daily activities. This research has the potential to greatly improve the usability

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and accessibility of technology, particularly for individuals with varying levels of visual impairment who need tailored navigation assistance.

Many applications can be envisaged beyond the use cases tested in the project. For example, solutions could be developed to assist VIPs in education, transport and mobility, tourism, public spaces, websites, etc. In education, one can think in assistive technologies for promoting inclusive learning environments. Similar applications can be envisaged in enterprises for assisting employees with disabilities. More broadly, in community programs that encourage social interaction and participation among people with disabilities.

The results of this work can impact Human Robot Interaction (HRI) in general, but particularly in unfamiliar environments where humans and robots must explore and navigate environments under tough conditions, such as natural disasters or war (e.g., HRI for rescuing people after an earthquake or an attack). In these situations, the communication wide-band is often limited, impacting the quality of the transmitted multimodal data. One potential solution is for humans and robots to engage in conversation to overcome these limitations. Furthermore, assistance robots could leverage on VIDI to navigate effectively and to proactively interact with victims by providing clear instructions, detecting dangers, and engaging in ongoing conversations to better meet the needs of those in distress.

b Scientific impact

This project pushes the limitations of HRI and current assistive technologies, by building VLMs capable of engaging in conversation with VIP, warning them about changes in the environment as well as providing navigation guidance. To accomplish this, the VLMs should support grounding, by conversing continuously with VI individuals, explicitly addressing misunderstanding as well as agreeing on what the system captures, what the VI person perceives and desires. It should also support planning to solve the guidance tasks by completing distinct intermediary sub-tasks. The scientific findings of this project can be used by other researchers to evaluate future models for assisting visually impaired individuals, notably through the dataset, the models and the evaluation benchmarks that we will provide.

c Valorization strategies

We are committed with academic research; thus, we plan to publish our findings in top conferences and journals across various fields including computer vision (CVPR, ICCV, ECCV), natural language processing (ACL, COLING, EMNLP, LREC), robot learning (CoRL, ICRA, IROS) and artificial intelligence (AAAI). For reproducibility reasons, we will ensure that the research finding will be accessible on public archives like HAL. We will advocate for open-source datasets and models to enhance reproducibility within the research community.

The results of this work can impact human-machine interactions, particularly for humans with limited vision or those needing hands-free devices, such as telecommunication technicians in the field requiring assistance or in-car assistance. Technicians can benefit from VIDI during intervention to have navigation support, contextual information, enhanced communication with teams or remote experts, potential hazards detection, etc. They could also leverage on VIDI as a real-time training tool, enabling them to adapt quickly to new situations. Handling multimodality at high speed is a challenge, VIDI's prototype will provide insights into the latency needs of real-time interactions.

Orange as the only industrial partner sees the opportunity to enhance its network and to develop inclusive services for employees, customers and its corporate clients. **By leveraging the prototype developed in this project, Orange Research will progressively transfer the innovation to delivery projects through other entities of the group such as Enovacom, which specializes in the healthcare domain, or the business units. This will enable the company to fulfill its commitment to digital accessibility.** Moreover, the transfer to services will open opportunities in network management and technical assistance.

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